

Jacob Rosario

Software Engineer

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Professional Summary

Early-career Software Engineer with hands-on experience building and deploying web and iOS applications using JavaScript, React, Swift, SwiftUI, Python, and Three.js. Developed real-time data and motion-sensing tools used by coaches and athletes across 20+ competitive matches. Skilled at turning complex data and user needs into responsive, practical software products.

Skills

- **Languages:** JavaScript, Python, Swift, HTML, CSS
- **Frameworks & Libraries:** React, SwiftUI, Three.js
- **Tools & AI:** Claude, ChatGPT, LLMs, Git, GitHub

Relevant Experience

Software Developer & Sports Analyst

Feb 2026 - May 2026

Hustle Volleyball Club

Burnaby & Vancouver, BC

- Developed and integrated a custom JavaScript stat-tracking application into match preparation and live coaching workflows, informing lineups, practices, and in-game adjustments
- Applied it across a 15-player competitive team's season of 5+ tournaments, including provincials and nationals.
- Software directly supported the team's qualification for the national championship through data-informed preparation.

Projects

RainCouver (Rain-Cover Navigation) | React, TypeScript, LiDAR/OSM data

Present

- Building a PWA that routes pedestrians through Downtown Vancouver by maximizing covered walking paths, using LiDAR-derived cover data across 25,000+ street segments and 3,000+ searchable places.
- Designing a self-improving routing system where user corrections update cover data in real time without disrupting an active route, solving GPS drift and ambiguous-turn edge cases along the way.

RepPower | Swift, SwiftUI [RepPower](#)

Sep 2026

- Developed an iOS application that uses AirPods motion-sensor data to estimate barbell squat velocity for velocity-based training.
- Engineered a Swift physics pipeline that transforms raw motion data into vertical acceleration and velocity while accounting for sensor noise and integration drift.
- Implemented zero-velocity drift correction and optimized high-frequency processing to maintain stable real-time measurements without blocking the SwiftUI interface.

Volleyball Stat Tracker | JavaScript, React [Volleyball-Stat-Tracker](#)

Feb 2026

- Deployed a real-time React app used courtside on iPad across 20+ competitive matches, replacing handwritten tracking with live performance data.
- Refactored redundant stat sources into a single source of truth, eliminating state drift and supporting tactical decisions that contributed to the team's Division 3 → Division 2 climb.

Spike Timer Analyzer | JavaScript, React [Spike-Timer](#)

Mar 2026

- Built a client-side video-analysis tool that measures spike timing from user-marked takeoff, contact, and landing frames.
- Selected a lightweight geometric approach instead of pose-detection machine learning, enabling responsive browser-based processing.

Other Experience

Performance Analyst & Assistant Coach

Sep 2025 - Dec 2025

Vancouver College

Vancouver, BC

- Analyzed 5+ matches/week using video review and statistical data to identify opponent tendencies and performance patterns.
- Translated video data into actionable recommendations for lineups, opponent-specific strategies, and player development.

Barista

Sep 2023 - Sep 2025

Starbucks

San Francisco, CA & Vancouver, BC

- Processed 100+ transactions per shift at high-volume locations while maintaining accuracy and service quality under pressure.

Education

University of British Columbia

Jun 2025

Bachelor of Science (BSc), Biology

Vancouver, BC

- **Achievements:** Dean's List

Certifications

- Anthropic | Claude Code 101, Claude 101, AI Fluency: Frameworks & Foundations

Hobbies & Interests

Competitive volleyball, coaching & performance analysis | Sports analytics & biomechanics | AR/VR & 3D technology | Video games | Weightlifting & Olympic lifting | Content creation & video editing | Anime & manga | Playing guitar